

# Collective thriving and team creativity in college students' innovation teams: A serial mediation model

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## ABSTRACT

The position of college students' innovation teams in college innovation education has been increasingly valued. Thriving may provide an important competitive advantage for team development; however, very few studies have examined the significance of collective thriving in college innovation teams. Based on theories such as the input-process-output (IPO) framework model and Chinese innovation teams as samples, this study aims to reveal how team voice and team mental model mediate the relationship between collective thriving and team creativity in a sample of Chinese innovation teams. A total of 223 innovation teams completed the online questionnaires. Positive correlations among collective thriving, team voice, team mental model, and team creativity were revealed. Moreover, team voice and the team mental model mediated the association between collective thriving and team creativity and served as serial mediators. This study elucidates how collective thriving influences college students' team creativity. Furthermore, the findings have important implications for promoting college students' team creativity.

## 1. Introduction

Innovative talent is the great driving force for social and economic development in the 21st century, and cultivating innovative talent is naturally the main task of education in every country worldwide. Innovation is an important driving force for a country's economic, social and cultural progress, and as the basic working unit of organizational innovation and an important force of the national innovation system, the creative ability of university research teams is directly related to the scientific development and technological innovation of the country (Zhou & Xin, 2023). At present, many colleges and universities in China are actively responding to the call for a national innovation construction strategy, seizing the opportunity to carry out system reform and path exploration for innovative talent training. As an effective form of talent cultivation, the status and role of college students' innovation teams in college innovation education has been increasingly valued. An innovation team comprises students who cooperate with each other to carry out innovative activities and produce valuable results to cultivate innovation ability and achieve learning goals or scientific research goals (Guo et al., 2008). College students have greater need for work placement autonomy and work interaction requirements than does any other group (Morgeson & Humphrey, 2006). From the 1980s to the present, researchers have conducted in-depth studies on team creativity in business organizations and the mechanisms that influence it, and many related studies have been carried out (Antonio et al., 2022; Wang et al., 2023). However, the findings on team creativity in corporate organizations are not

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necessarily applicable to university student innovation teams. Therefore, this study focuses on how to promote team creativity among university students.

Thriving represents a positive mental state of learning and vigor, where learning refers to the acquisition of knowledge and skills and the ability to use them, and vigor is a positive attitude toward energy (Spreitzer et al., 2005). Positive organizational behavior theory emphasizes the development and management of individual strengths and potentials and considers how organizations can take positive steps to leverage the strengths of their employees to improve individual, group and organizational performance, focusing on psychological states that can be changed by the individual, which should be effectively measured and malleable (Campbell-Quick et al., 2010; Luthans, 2002). This study focused on the psychological state of thriving, and thriving is important for both individuals and organizations. Researchers continue to explore the impact of individual positive states (thriving) at different levels, both the individual and team levels (Yammarino et al., 2008). Thriving is inherently multilevel, but much of the existing scholarship locates prosperity at the individual level to explain differences in individual prosperity; however, the same explanation may not explain differences at the team level (Goh et al., 2022). This can lead to false generalizations or fallacious levels, which can hinder structural clarity and the ability to thrive on theoretical advances. Collective thriving has not received enough attention in current research (Han et al., 2022). As a result, there is a call for research on the collective level of thriving and a general consensus that thriving may provide an important competitive advantage for team or organizational development (Kleine et al., 2019; Spreitzer & Sutcliffe, 2007); to illustrate this point, the present study will reveal the effects of collective thriving on college student team creativity to shed light on college student team creativity formation.

According to the input-process-output (IPO) framework model (McGrath et al., 2000), a team's results are affected by the team's input factors through interactions between members. The cognitive, motivational, and affective states of the team are related to the nature of member interactions but do not describe the nature of member interactions and are therefore considered team input factors (Marks et al., 2001). Collective thriving encompasses two factors, learning and vigor, with learning being a team cognitive factor and vigor being a positive affective factor (Zheng & Lu, 2013). According to the I-P-O model, collective thriving is a team input factor. Can collective prosperity on the input side affect team creativity through interaction and sharing? Team voice and team mental model involve team interactions and are considered process factors (Cohen & Bailey, 1997); team creativity is considered a team output factor (Li et al., 2024). Thriving teams value collective discourse, including frequent sharing, communication, listening, and questioning (Narel et al., 2019). The voice behaviors of team employees can facilitate information exchange, which is conducive to team creativity (Deng et al., 2018; Jian & Chen, 2023). In addition, according to broadening-constructing theory, people's positive emotions (e.g., vigor) broaden their attention and thinking and construct long-term resources (Fredrickson & Branigan, 2005), and learning is the main source of team members' knowledge stock and sharing (Ma & Yu, 2019). However, team innovation also requires the development of shared knowledge structures through interactions between team members, as well as accurate interpretations and expectations of the task, i.e., the team mental model. However, few integrative studies have explored the joint roles of team voice and team mental model), revealing much less about the pathways through which the two play a role in collective thriving affecting college students' team creativity.

With this in mind, this study sought to explore the mechanisms through which collective thriving affects college student team creativity and to examine the mediating role of team voice and team mental model in this process, thereby enriching the literature on team-level creativity and enhancing our understanding of how and why collective thriving affects the field of college student team creativity.

## 2. Literature review

### 2.1. Collective thriving and team creativity

Creativity refers to the generation of ideas (Anderson et al., 2014). Team creativity is the development of appropriate and novel ideas, solutions, or processes in the pursuit of team goals (Amabile, 1996). Meta-analytical studies have shown that team-level factors have a key influence on creativity (Wang et al., 2011). Collective thriving is a positive collective state of mind in which the members of a collective can learn and benefit in a way that meets the needs of the collective, fostering collective competence and collective development in response to unpredictable risks and challenges (Spreitzer & Sutcliffe, 2007). Groups and organizations experience collective thriving and upward mobility, which in turn produces good outcomes at the group and organizational levels (Spreitzer et al., 2005).

Previous empirical studies have supported the impact of thriving at work on individuals' innovative behavior (Alikaj et al., 2021; Liu et al., 2020; Riaz et al., 2020). When individuals experience learning and vigor, they tend to identify problems and come up with new ideas, and they are more motivated and energetic to engage in innovative activities (Carmeli & Spreitzer, 2009; Khurshid et al., 2021). However, few empirical studies have explored the relationship between collective thriving and team creativity, and there is a lack of findings in the academic field. Groups, teams, and organizations all experience collective thriving upwards, which in turn produces good results at the group and organizational levels (Spreitzer & Sutcliffe, 2007). Given that team creativity is becoming an important component of innovation education in colleges and universities (Zhang et al., 2023), the relationship between collective thriving and team creativity in the academic field is equally worth exploring. Innovation teams of university students in higher education are generally self-organized by students with a relaxed learning atmosphere, strong interests and a clear division of labor and consistent learning or research goals (Cao & Tang, 2023; Hülshager et al., 2009), with members mostly focusing on self-development and transcendence and tending to engage in more knowledge sharing (Buch et al., 2017; Nerstad et al., 2018). In contrast, members of corporate innovation teams may face internal competitive pressures, work-family role conflicts, and other pressures, which can hinder

their sharing behaviors within the team (Ding & Wang, 2023; Wang et al., 2022;) and prevent employees from presenting a vibrant work state (Zhou et al., 2023); moreover, reducing knowledge sharing will reduce team members' ability to generate creative ideas (Bartol et al., 2002). In addition, students and student innovation teams are more creative at learning scientific knowledge and solving scientific problems than corporate employees and research and development teams are (Yi & Hu, 2013). Creativity requires knowledge-based abilities and skills to address new things or ideas, and learning and developing new knowledge and skills are essential for creativity (Amabile, 1983). Thriving teams are able to build collective capabilities and new competency traits through learning, and these collective capabilities can meet an unpredictable world (Campbell-Quick et al., 2010). Thus, a thriving team can be associated with broader information processing strategies and creative generation (Fredrickson & Losada, 2005). In addition, a thriving collective is energized, which helps the collective cope with challenges, obstacles, and setbacks and persevere (Campbell-Quick et al., 2010). Thriving teams build new competencies through learning; improve the collective ability to cope with setbacks, challenges, and failures; and persevere in their endeavors through the power of dynamism (Spreitzer & Sutcliffe, 2007). Thus, creativity requires not only that teams continue to learn and identify problems but also that members have sufficient energy to overcome challenges. The following hypotheses are proposed:

Hypothesis 1: Collective thriving is positively related to team creativity.

## 2.2. Mediating role of team voice

Team voice is the extent to which members make constructive suggestions for improvement, share new ideas, and discuss potential problems or obstacles (Walumbwa et al., 2012). It is the extent to which members collectively express concerns, suggestions, and ideas about work-related issues (Li et al., 2017). Team voice is one of the key conditions of team creativity (Ye et al., 2019). Considering the IPO model, Marks et al. (2001) explained the process factors affecting team creativity as interdependent activities that coordinate task work. Thus, collective thriving may act on team creativity through team voice.

Thriving encourages people to engage in behaviors that promote long-term resource development, and team voice can be seen as a positive means of building long-term resources for future development in several ways (Sheng & Zhou, 2021). First, the learning factor in thriving can expand horizons and enrich the content of voice (Koçak & Agun, 2019), which encourages individuals to identify problems and propose better solutions (Carmeli & Spreitzer, 2009). Second, the vitality factor of thriving at work can enhance individuals' confidence and courage and provide energy for voice (Koçak & Agun, 2019). This promotes the sharing and implementation of new ideas among individuals (Elahi et al., 2020) and goes beyond individuals' established responsibilities and roles (Yousaf et al., 2019). However, the research findings on individual thriving cannot explain the state of the collective (Spreitzer & Sutcliffe, 2007). Researchers believe that thriving teams attach importance to the right to speak collectively (frequent sharing, communicating, listening, and questioning) and have the cultural characteristic of continuous improvement (Narel et al., 2019). College student innovation teams are structurally and intensively interactive and communicate and learn from each other, and a vibrant positive affective state is also generated during communication and interaction (Campbell-Quick et al., 2010). According to the motivational information processing model (MIP-G), when collaborators are aware of the differences in views and information in the group, their awareness of learning increases, and they are more likely to search for, share and consider new information (De Dreu et al., 2008). The members of college students' innovation teams come from different colleges and majors, and differences in information make them more knowledgeable about learning and sharing new knowledge and ideas. A thriving team consists of members who are both energetic and eager to learn new things, which allows them to find new problems and formulate solutions to existing problems together (Wu & Chen, 2019). Therefore, thriving at the collective level may positively impact team voice. Furthermore, Bledow et al. (2009) suggested that team creativity must be based on enabling different opinions and perspectives within the team. As the team is created to stimulate the participation of each member, creativity requires members to voice their own views and opinions. Moreover, the influence of team voice runs through the entire team creativity process. During the stage of team creativity generation, team creativity is promoted when members propose different ideas that challenge the current situation and when they implement useful ideas (Liang et al., 2019). Moreover, team innovation is an unusual behavior that implies the rejection or modification of previously accepted thoughts beyond conventional assumptions (Eisenbeiss et al., 2008). In the implementation stage of team creativity, voice can encourage members to engage in divergent thinking, form broader strategic alternatives, and carefully consider the feasibility of these alternatives (Morrison, 2011), which promotes the successful implementation of team creativity. Furthermore, when members make their voices heard, the team can use these inputs to replace original practices with better practices (Li et al., 2017). The following hypothesis is proposed:

Hypothesis 2: Team voice mediates the relationship between collective thriving and team creativity.

## 2.3. Mediating role of team mental model

The team mental model reflects the team's "shared" cognition; its purpose is to analyze, explain, and predict the surrounding environment (Carley, 1997; Liu et al., 2017). This model can reflect implicit coordination in an effective team and can help team members understand how the team operates in an uncertain and dynamic environment (Mohammed et al., 2010). Furthermore, it promotes the adoption and coordination of views among members and impacts creativity (Toader & Kessler, 2018). According to the IPO model (McGrath et al., 2000), team input factors (e.g., collective thriving) may affect team creativity through a team mental model.

The broaden-and-build theory offers an explanation for why greater collective thriving strengthens the team mental model (Fredrickson & Branigan, 2005). When a team experiences a sense of vitality from work, it is usually accompanied by a positive experience that can expand team cognition. For example, an organizational culture that attaches importance to vitality can promote members' exploration and innovation, positively impacting implicit knowledge sharing (Ji & Wang, 2012). Furthermore, the learning characteristics of thriving teams continue to enrich their mental model. Zheng and Lu (2013) suggested that individuals with a strong sense of vitality can bring new knowledge and ideas into the organization through learning, which can provide more knowledge reserves for the organization. Vigorous teams are characterized by information sharing (Sonenshein et al., 2006); thus, they can quickly identify, coordinate, and implement actions when problems arise. Therefore, collective thriving may positively predict a team's mental model. Furthermore, some studies have shown that teams' mental model can promote effective coordination and cooperation among team members, thus stimulating creativity (e.g., Li & Xu, 2013; Santos et al., 2015). Many researchers agree that if members share similar ideas, they will not discuss different views, which inhibits creativity (e.g., Skilton & Dooley, 2010; Toader & Kessler, 2018). College student innovation team members come from different majors and possess different professional backgrounds, knowledge and skills. When confronted with complex task contexts and requirements, effective interactions among members can stimulate each other to think about and integrate the different perspectives and viewpoints to which they are exposed, leading to novel ideas (Perry-Smith & Shalley, 2003). Other studies suggest that there is an inflection point in the association between team mental model and creativity, during which individuals exhibit a trend of first rising and then falling (e.g., Huang & Peng, 2015). Nevertheless, it remains unclear how team mental model impact team creativity. When members have a team mental model, the interpretations and expectations formed by tacit understanding will enable them to adjust their daily work in complex and dynamic task environments (Resick et al., 2010). In addition to a common knowledge structure among team members, the team mental model also serves as the guiding belief for teamwork. A team mental model is conducive to communication and cooperation, which leads to creativity and subsequently improved innovation performance (Ji & Cai, 2013). Through these functions, members of a team mental model can develop new ideas or products that are consistent with task requirements and team needs (Hülshager et al., 2009). The following hypothesis is proposed:

Hypothesis 3: A team's mental model mediates the relationship between collective thriving and team creativity.

#### 2.4. Serial mediating role of team voice and team mental model

According to the cognitive theory of team interaction, team cognition depends heavily on the coordination of team members' interactions in a task situation (Cooke et al., 2013), and social information processing theory suggests that social interactions can facilitate shared cognition among team members (Pfeffer & Salancik, 1977). These findings suggested that team voice may influence a team's mental model. According to previous research, members present their own ideas and opinions, which provides team members with the opportunity to share knowledge (Kurtzberg & Amabile, 2001). Voices of any nature promote the integration of diverse perceptions and expand team perceptions (Taggar, 2002). Therefore, the more open the communication between team members is, the easier it is to form a shared mental model (Wu & Wu, 2005). In the complex and uncertain task environments faced by university student innovation teams, team members must maintain task-oriented interactive communication, which facilitates consistent cognition and promotes task knowledge coordination (Hülshager et al., 2009). A key condition for knowledge sharing is that members fully and accurately understand each other's perspectives and knowledge. This study demonstrated that the combination of team members' voice and ideas creates tacit understanding and that a shared knowledge structure contributes to the achievement of team goals. Team voice can generate new ideas by facilitating information and knowledge sharing and communication conflicts (Ye et al., 2019). Teams that exhibit active voice behaviors have more comprehensive and rich knowledge and information, deeper interactions, and greater creativity levels than other teams (Deng et al., 2018). Compared to silent teams, teams that speak freely are more likely to develop an atmosphere of positive voice, deepen their understanding of team goals and tasks, and form an identity with the team (Zhu & Wang, 2019). Therefore, the team mental model is a key factor in team creativity. The development of a high-quality mental model requires reliance on team voice, and collective thriving creates positive conditions for the development of team voice. The following hypothesis is proposed:

Hypothesis 4: Collective thriving can affect team creativity through the serial mediating roles of team voice and team mental model.

In summary, this study systematically explored the association between collective thriving and college students' team creativity and the mediating roles of team voice and the team mental model in the above association (Fig. 1). Finally, this study reveals how

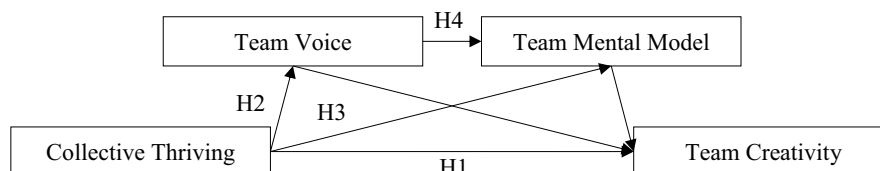


Fig. 1. Research model.

collective thriving affects college students' team creativity.

### 3. Methods

#### 3.1. Participants and procedure

Using an online questionnaire platform to collect data, we conducted a survey on team creativity among university students who applied for and participated in the China University Innovation and Entrepreneurship Training Program. The team members come from different specialties and volunteer to form teams according to their interest in conducting exploratory research on innovative research projects for a year of collaboration. The innovation teams that this study focuses on apply, develop, and implement innovation projects autonomously for a limited period under the guidance of an instructor to achieve innovation in knowledge, technology, and research methods. University student innovation teams utilize experimentation as a means and innovation as an end, focusing on innovation and collaboration to solve problems in the discipline, across disciplines, and within the enterprise R&D process. Teams are independent of the creation and division of labor and have strong independent thinking abilities and senses of innovation. The teachers responsible for the application of the innovation training project helped us distribute the questionnaires to the WeChat group of the university student innovation training project (group members included team leaders, instructors, and other leaders). These teams had existed for approximately 1 year. The participants volunteered to participate in this survey and received a small gift. The participants remained anonymous and could withdraw at any time. The participants could complete the questionnaire only after all the items were completed. The Henan Normal University Ethics Committee supervised our study.

We collected 992 questionnaires in total. For 57 questionnaires, the answers to all questions were the same or followed a similar pattern; thus, they were removed. Questionnaires that were clearly outside the normal time range at which respondents responded were deleted; therefore, 27 invalid questionnaires were excluded because their response time was longer than 660 s (6 invalid questionnaires) or less than 60 s (21 invalid questionnaires). After deleting the above invalid questionnaires, if only one or two members of a team were left to complete the questionnaire, the questionnaire (59 invalid questionnaires) was removed. In preparation for the coming year's project, informal members joined the team for learning purposes and undertook some of the team's minor tasks (e.g., personnel contact, academic material collection). The relative lack of time for team collaboration may have reduced the objectivity of the data, so data from informal team participants (16 students) were removed.

After eliminating invalid questionnaires, 833 valid questionnaires remained. A total of 546 women (65.5 %) and 287 men (34.5 %) were included. Patient ages ranged from 23 to 24 years. These samples formed 223 teams. A total of 110 (49.3 %) teams responded with 3 people, 61 (27.4 %) with 4 people, and 52 (23.3 %) with 5 people. The average team response rate (the number of questionnaires obtained within a team/team size) was 95.2 %. There were 181 (81.2 %) teams with a 100 % response rate, 2 (0.9 %) with an 80 % response rate, 38 (17.0 %) with a 75 % response rate, and 2 (0.9 %) with a 60 % response rate. The size of all the teams was 3–5 people. There were 70 (31.4 %) three-person teams, 97 (43.5 %) four-person teams and 56 (25.1 %) five-person teams. The average team size was 3.9 people. Moreover, there were 64 teams (28.7 %) with a male ratio greater than 75 %, 26 (11.7 %) with a male ratio of 25 %–75 %, and 133 (59.6 %) with a male ratio less than 25 %.

#### 3.2. Measures

Collective thriving was measured using a 10-item scale validated by [Porath et al. \(2012\)](#). Chinese researchers have widely used the Chinese version of the scale to evaluate thriving (e.g., [Wang & Wei, 2021](#)). Items were rated on a 5-point scale from 1 (strongly disagree) to 5 (strongly agree). The scale included two subscales: learning latent factors (such as “We are not learning when working in the team”) and vitality latent factors (such as “We have energy and spirit when working in the team”). The Cronbach's alpha in our study was 0.893. CFA ( $\chi^2/df=3.853$ ,  $CFI=0.971$ ,  $TLI=0.963$ ,  $RMSEA=0.059$ ) was within acceptable limits ([Wen et al., 2004](#)).

Team voice was measured using the Chinese version of a 10-item scale developed by [Liang et al., \(2012\)](#). It was found that this scale was reliable when used to measure the voice of participants from China (e.g., [Long & Chen, 2021](#); [Zhang et al., 2020](#)). Items were rated on a 5-point scale. This scale included two subscales: promotive voice (e.g., “Raise suggestions to improve the team's working procedure”) and prohibitive voice (e.g., “Advise other members against undesirable behaviors that would hamper team performance”). Cronbach's alpha in our sample was 0.803. CFA ( $\chi^2/df=3.416$ ,  $CFI=0.956$ ,  $TLI=0.943$ ,  $RMSEA=0.054$ ) was within acceptable limits.

The team mental model was measured using a 6-item scale developed by [Cannon-bowers et al. \(1991\)](#). Previous studies in China have largely adopted the Chinese version of the scale, which is highly reliable (e.g., [Liu et al., 2017](#)). Items were rated on a 5-point scale. The scale included two subscales: an interactive mental model (e.g., “Whenever I need it, team members will reach out to me and offer help or advice”) and a teammate mental model (e.g., “Team members will adapt their own behavior to meet the needs of other members”). Cronbach's alpha in our sample was 0.835. CFA ( $\chi^2/df=3.055$ ,  $CFI=0.989$ ,  $TLI=0.981$ ,  $RMSEA=0.050$ ) was within acceptable limits.

Team creativity was measured using a scale developed by [Welbourne et al. \(1998\)](#). The Chinese version of the scale was reported to be highly reliable when assessing team creativity and innovation among Chinese participants (e.g., [Luo et al., 2017](#)). The scale was also included in the Handbook of Chinese Scales for Management Research (2nd edition) edited by [Li et al. \(2020\)](#). The scale was rated on a 5-point scale and had four items (e.g., “Team members can implement new ideas smoothly”). Cronbach's alpha in our sample was 0.878. CFA ( $\chi^2/df=3.330$ ,  $CFI=0.997$ ,  $TLI=0.992$ ,  $RMSEA=0.053$ ) was within acceptable limits.

The maleratio, team size, and number of team responses (abbreviated as team response) were selected as control variables during the data analysis. The proportion of males cannot be ignored in research on the collective scientific creativity of teams (Zhao et al.,



2021). There are differences in information acquisition and interpretation, cognitive perspectives and abilities, and the integration of interteam interactions between team members of different genders (Wu et al., 2021). Moreover, team size is closely related to cooperation and efficiency (Li et al., 2017). A team's size determines the range of situations in which processes can be embedded within the team and has a direct effect on team effectiveness (Mathieu et al., 2019). Team response is the number of questionnaires obtained from a team. Team response is related to team size, which may affect the validity of the data and reflects the consistency of team members' participation in the study.

The reliability and validity test results of the four scales are shown in Table 1.

### 3.3. Data analysis

A common-method bias test of the variables was performed using Mplus, and SPSS was selected for reliability analysis and correlation analysis. The Rwg (within-group interrater reliability) and ICC (intraclass correlation coefficient) were calculated by SPSS to check whether the aggregation constructed at the team level was reasonable. Finally, the hypotheses were examined using Mplus (5000 bootstrap resamples).

## 4. Results

### 4.1. Common method bias test

An unmeasured latent method factor was used to control for common method bias. All identification variables were allowed to load on the latent variable that represents common method bias in a structural equation model. Comparing the fit of the two cases with and without the latent variable allowed us to assess the common method bias effect. A common method variance factor was added to the four-factor model without significantly changing the main fit indicators ( $\Delta\chi^2=55.533$ ,  $\Delta df=1$ ,  $\Delta CFI=0.006$ ,  $\Delta TLI=0.007$ ,  $\Delta RMSEA=0.003$ ). In conclusion, common method bias did not significantly affect the variables after adding the common method bias factor.

CFA was used to test the structural validity between the variables. According to the results (Table 2), the four-factor model had high structural validity due to a better model fit ( $\chi^2/df=1.723$ ,  $CFI=0.968$ ,  $TLI=0.966$ ,  $RMSEA=0.029$ ) than the alternative models.

### 4.2. Data aggregation test

Because team-level variables were derived at the individual level, determining whether it was feasible to aggregate individual-level variables at the team level was necessary. A higher level of data aggregation is usually justified by the RWG and ICC (Newman & Sin, 2009). The Rwg mean values for collective thriving, team voice, team mental model, and team creativity (0.98, 0.98, 0.97, and 0.95, respectively) were greater than the cutoff points; thus, they met the aggregation requirements. The ICC (1) values of the four scales (collective thriving, team voice, team mental model, and team creativity) were 0.34, 0.39, 0.27, and 0.27, respectively, and the ICC (2) values of the four scales were 0.66, 0.70, 0.58 and 0.59, respectively. All the variables were within a significant range. Accordingly, the four individual-level variables can be aggregated to the team level.

### 4.3. Descriptive statistics and correlations between primary variables

Table 3 shows that collective thriving was correlated with team voice ( $\gamma=0.490$ ,  $p<0.01$ ), team mental model ( $\gamma=0.610$ ,  $p<0.01$ ), and team creativity ( $\gamma=0.565$ ,  $p<0.01$ ). Team creativity was correlated with team voice ( $\gamma=0.609$ ,  $p<0.01$ ) and team mental model ( $\gamma=0.595$ ,  $p<0.01$ ). Moreover, team voice was correlated with team mental model ( $\gamma=0.554$ ,  $p<0.01$ ). In conclusion, the statistical requirements were met for further hypothesis testing.

### 4.4. Hypothesis testing

Based on the correlation analysis, we tested how collective thriving and team creativity are mediated by team voice and team mental model (Fig. 2).

The results showed that the data and the model fit well ( $\chi^2/df=1.558$ ,  $CFI=0.937$ ,  $TLI=0.932$ ,  $RMSEA=0.050$ ) and indicated that team creativity was significantly impacted by collective thriving. In the path "collective thriving  $\rightarrow$  team voice  $\rightarrow$  team creativity," team voice was significantly impacted by collective thriving ( $\beta=0.562$ ,  $p<0.001$ ), while team creativity was significantly impacted by

**Table 1**  
Reliability and validity values.

| Variables           | Cronbach's alpha | $\chi^2/df$ | CFI   | TLI   | RMSEA |
|---------------------|------------------|-------------|-------|-------|-------|
| Collective thriving | 0.893            | 3.853       | 0.971 | 0.963 | 0.059 |
| Team voice          | 0.803            | 3.416       | 0.956 | 0.943 | 0.054 |
| Team mental model   | 0.835            | 3.055       | 0.989 | 0.981 | 0.050 |
| Team creativity     | 0.878            | 3.330       | 0.997 | 0.992 | 0.053 |

**Table 2**Comparison of the measurement models ( $n = 833$ ).

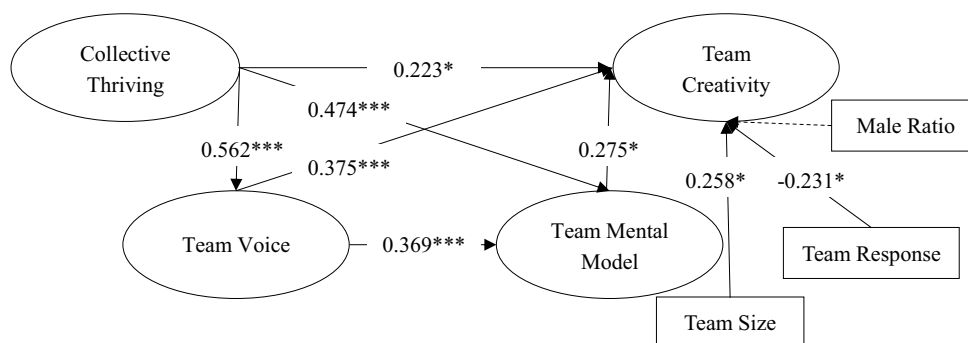
| Measurement models                  | $\chi^2/df$ | CFI   | TLI   | RMSEA |
|-------------------------------------|-------------|-------|-------|-------|
| Four-factor model 1: CT, TV, TM, TC | 1.723       | 0.968 | 0.966 | 0.029 |
| Three-factor model 2: CT, TV+TM, TC | 4.677       | 0.838 | 0.825 | 0.066 |
| Three-factor model 3: CT+TV, TM, TC | 5.769       | 0.791 | 0.773 | 0.076 |
| Two-factor model 4: CT+TV, TM+TC    | 9.482       | 0.626 | 0.597 | 0.101 |
| One-factor model 5: CT+TV+TM+TC     | 12.721      | 0.481 | 0.443 | 0.119 |

CT = collective thriving; TV = team voice; TM = team mental model; TC = team creativity.

**Table 3**Intercorrelations for variables ( $n = 223$ ).

|      | $M \pm SD$        | Min  | Max  | 1      | 2       | 3       | 4       | 5       | 6       | 7 |
|------|-------------------|------|------|--------|---------|---------|---------|---------|---------|---|
| 1MR  | 0.343 $\pm$ 0.440 | 0.00 | 1.00 |        |         |         |         |         |         |   |
| 2 TS | 3.937 $\pm$ 0.751 | 3.00 | 5.00 | 0.042  |         |         |         |         |         |   |
| 3 TR | 3.740 $\pm$ 0.814 | 3.00 | 5.00 | 0.027  | 0.858** |         |         |         |         |   |
| 4 CT | 4.073 $\pm$ 0.338 | 2.97 | 4.98 | 0.162* | 0.097   | 0.187** |         |         |         |   |
| 5 TV | 3.978 $\pm$ 0.294 | 3.17 | 4.87 | 0.072  | 0.054   | 0.090   | 0.490** |         |         |   |
| 6 TM | 3.966 $\pm$ 0.376 | 2.94 | 4.78 | 0.078  | 0.096   | 0.132*  | 0.610** | 0.554** |         |   |
| 7 TC | 4.200 $\pm$ 0.463 | 2.58 | 5.00 | 0.119  | 0.096   | 0.082   | 0.565** | 0.609** | 0.595** |   |

MR = Male ratio; TS = Team size; TR = Team response; CT = collective thriving; TV = team voice; TM = team mental model; TC = team creativity.

\*\*  $p < 0.01$ .\*  $p < 0.05$ , two-tailed.**Fig. 2.** Serial mediated effects model estimation results (standardization).

team voice ( $\beta=0.375$ ,  $p<0.001$ ). Thus, collective thriving enhances team creativity by enhancing students' team voice; thus, Hypothesis 2 was supported. In the path "collective thriving  $\rightarrow$  team mental model  $\rightarrow$  team creativity," the team mental model was significantly impacted by collective thriving ( $\beta=0.474$ ,  $p<0.001$ ), while team creativity was significantly impacted by the team mental model ( $\beta=0.275$ ,  $p<0.05$ ). Thus, collective thriving enhances team creativity by enhancing students' team mental model; thus, Hypothesis 3 was supported. In the path "collective thriving  $\rightarrow$  team voice  $\rightarrow$  team mental model  $\rightarrow$  team creativity," the team mental model was significantly impacted by team voice ( $\beta=0.369$ ,  $p<0.001$ ). Thus, the promotion of team voice was helpful in the formation of students' team mental model. This means that collective thriving enhances team mental model by improving team voice, thereby enhancing student team creativity; thus, Hypothesis 4 was supported. Furthermore, team creativity was significantly impacted by collective thriving ( $\beta=0.223$ ,  $p<0.05$ ); thus, Hypothesis 1 was supported. Overall, team voice and the team mental model had partial mediating effects on the relationship between collective thriving and college students' team creativity.

**Table 4**Effects and 95 % CI for the mediation model ( $n = 223$ ).

| Model pathways                                                                                                                | Standardized Effect | Percentage of Effect | Boot SE | 95 % CI LLCI | ULCI  |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|---------|--------------|-------|
| Direct effects: Collective thriving $\rightarrow$ Team creativity                                                             | 0.223               | 36.00%               | 0.091   | 0.045        | 0.401 |
| Indirect effect 1: Collective thriving $\rightarrow$ Team voice $\rightarrow$ Team creativity,                                | 0.210               | 33.87 %              | 0.060   | 0.093        | 0.328 |
| Indirect effect 2: Collective thriving $\rightarrow$ Team mental model $\rightarrow$ Team creativity                          | 0.130               | 21.00 %              | 0.054   | 0.024        | 0.236 |
| Indirect effect 3: Collective thriving $\rightarrow$ Team voice $\rightarrow$ Team mental model $\rightarrow$ Team creativity | 0.057               | 9.19 %               | 0.024   | 0.011        | 0.103 |

To test the serial mediation model, the bias-corrected percentile bootstrap method was applied (Table 4). The 95 % CI (confidence interval) for the direct predictive effect of collective thriving on team creativity was [0.045, 0.401]. The direct path was significant, and the percentage of effect was 36.00 %; thus, Hypothesis 1 was supported. The estimated mediating effect of team voice between collective thriving and college students' team creativity was 0.210 (95 % CI=0.093, 0.328). Mediated path 1 was significant, with a percentage of effect of 33.87 %, and Hypothesis 2 was supported. The estimated mediating effect of the team mental model between collective thriving and college students' team creativity was 0.130 (95 % CI=0.024, 0.236). Mediated path 2 was significant, with a percentage of effect of 21.00 %, and Hypothesis 3 was supported. The serial mediated effect estimate for team voice and team mental model between collective thriving and college students' team creativity was 0.057, with a 95 % CI [0.011, 0.103]. Mediated path 3 was significant, with a percentage of effect of 9.19 %, and Hypothesis 4 was supported. Collective thriving mainly improved team creativity by enhancing team voice.

## 5. Discussion

This study investigated the mechanism through which collective thriving influences team creativity among college students. Collective thriving was positively correlated with team creativity; collective thriving and team creativity were mediated by team voice and team mental model, and these variables were serially mediated. College students on innovation teams with high collective thriving styles reported more team voice, which further enhanced the construction of the team mental model and promoted team creativity.

### 5.1. Theoretical implications

This study enriches the research content on team creativity and expands the research on prosperity outcomes.

In this study, college students' team creativity was positively impacted by collective thriving. This finding suggests that the positive state of a team is indeed an important factor in inspiring team creativity. Several researchers have suggested that the association between thriving and team results should be examined (Kleine et al., 2019). This study followed this call and showed that the conclusion at the team level was in agreement with the conclusions of research conducted at the individual level: creativity is driven by a perception of learning and vitality (e.g., Khurshid et al., 2021; Wang et al., 2021). The possible reasons for these findings are as follows. According to the perspective of the broaden-and-build theory (Fredrickson & Branigan, 2005), by sharing vitality, teams can take more actions or behaviors to accomplish tasks, thus creating more opportunities for creativity. Group dynamic theory (Lewin, 1947) emphasizes the examination and study of group behavior from the perspective of group endogeneity. Perceptions of the group and of the work it performs have multifaceted effects on the direction, intensity, and other aspects of group behavior, and group morale is a feeling of satisfaction and vitality toward the group and is necessary to achieve group goals. As a result of this research, we found that collective thriving, which includes both learning and vitality latent factors, is positively correlated with team creativity performance, further extending group dynamic theory. Furthermore, thriving teams have a generative, interdependent, emotional environment and high team adaptability; thus, team members tend to help each other (Walumbwa et al., 2018) and exhibit cooperative, exploratory, and focused behaviors (Keister, 2014). Therefore, collective thriving can create stronger relationships among team members and increase enthusiasm for team cooperation, creating positive conditions for team creativity.

This study investigated the association between collective thriving and team creativity among Chinese innovation teams comprising college students. Research on the impact of cultural context on individual thriving is encouraged (Shahid et al., 2021; Walumbwa et al., 2018). Furthermore, creativity is closely related to culture (Li et al., 2020). Unlike the Western individualistic approach to interpersonal relationships, "relationships" can reflect the distinctive characteristics of the Chinese context, so the interaction process cannot be ignored when studying thriving (Yao & Li, 2021). Confucian culture has long influenced Chinese culture (Zhang et al., 2022) and has its own unique views on education and learning, such as learning for pleasure and shared learning. These Chinese cultural characteristics may affect collective thriving to some extent. To this end, previous individual-level studies on the association between thriving and creativity have considered Chinese culture (Wang et al., 2021). When it is expressed as a higher-level phenomenon through interactive amplification, the team will rise to a collective state (Kozlowski & Klein, 2000). That is, the interconnected nature of the team and the formation of a specific working group form a symbolic situation in which the collective state is formed (Curşeu, 2006). It is therefore appropriate to examine the collective state of thriving within the team as a unit of analysis. However, the role of thriving at the collective level has been examined in few studies of team creativity, particularly in the Chinese context. Furthermore, most studies on the association between thriving and other variables have focused on employees' thriving in the workplace (Porath et al., 2012). However, the research findings on enterprise research or development teams may not be applicable to college students' innovation teams. Considering Chinese culture, the impact of collective thriving on voice and mental model in influencing college students' team creativity is presented in this study. These findings have positive implications. Thus, our study offers a theoretical framework for the management of innovation teams of college students in other countries and a foundation for studying different cultures and cross-level thriving. It integrates two mediating mechanisms of team voice behavior and team mental model.

In the framework of the IPO model, this paper proposes a series of mediations, team voice and team mental model. In this study, team voice and team mental model were demonstrated to be mediating factors in the association between collective thriving and college students' team creativity. Serial mediation was also found in the above relationship. Thus, the internal mechanism through which collective thriving affects team creativity was explained, which not only deepens the internal logic of the IPO model but also fills the theoretical gap between collective thriving and team creativity. Furthermore, this study showed that collective thriving positively affects team creativity and helps improve team voice and mental model. Therefore, our study extends the limited literature on collective thriving. As outlined in a study of group functions (Yuan, 2018), the purpose of active groups is to fulfill organizational



objectives and to meet the needs of their members for communication, cooperation, and knowledge sharing. The potential explanations for these findings are as follows. Team members who experience collective thriving have more identification, emotional attachment, and participation in their team (Walumbwa et al., 2018), demonstrating the group's proactive behavior (Keister, 2014). A thriving team is characterized by a culture that values continuous advice, ensuring that each member has the right to make suggestions and promote communication and sharing (Narel et al., 2019). Furthermore, collective thriving enables team members to quickly identify and coordinate opinions and behaviors when problems arise (Sonenshein et al., 2006) and connects the team with broader ideas (Fredrickson & Losada, 2005). By participating in team voice, everyone on the team is encouraged to form new connections, determine what works and what does not, and integrate different ideas in a creative manner. It is also conducive to the formation of tacit understanding, provides complementary advantages, facilitates the development of a shared mental model, and promotes the generation of follow-up ideas. Similarly, Wang and Xue (2013) suggested that the key to team creativity was to acquire and integrate members' unique knowledge and abilities. Overall, collective thriving is beneficial for team creativity by promoting team voice and fostering a team mental model.

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## 5.2. Practical implications

Our findings have important implications for team creativity. First, a team needs to enhance collective thriving to show a positive state of team creativity. The socially embedded model of thriving at work states that thriving takes place in a stable working environment or in one with active resources (Spreitzer et al., 2005). The team should create a working atmosphere of trust and respect as much as possible and allow members to have free decision-making power. With respect to resources, teams should provide team members with rich innovation resources and learning materials, enable an independent innovation space, and create harmonious interpersonal relationships within the team. Second, a team should actively listen to individual voices to increase its innovation interaction. In addition, team members may not feel that their opinions are valued or expected by others, which can hinder team voice (Ye et al., 2019). When members think their opinions are valued by the team, they are likely to have an enhanced sense of self-worth in expressing views. Therefore, in innovation practice education, teams should create a free and inclusive team atmosphere, unblock rationalization suggestion channels, and encourage team members to strengthen their beliefs and speak freely in any situation. Finally, teams should strengthen their mental model to consolidate their innovation and sharing power. During innovation, teams should actively build a shared team mental model by paying attention not only to the disciplinary knowledge and innovation skills of members but also to the complementarity of expertise, styles, methods, and information among members. This will promote the maturity of the team mental model and enrich university students' network of innovation team knowledge. In addition, the team must encourage members to establish a tacit understanding to jointly achieve the objectives of team cooperation and resource coordination by providing stable support for the generation and implementation of innovation.

## 5.3. Limitations and future directions

Although this study provides useful information for research on collective thriving and team creativity, several limitations should be noted. First, while the team creativity scale, which has only 4 items, has some scientific validity, it lacks many statistical results as an empirical basis, and the small number of items might influence the findings. Therefore, it may be necessary to select a scale that has more items for retesting to improve the explanatory strength of the findings. Second, all the assessments were completed by the team members via self-reports. Conclusions regarding creativity derived from self-perceptions should be interpreted with caution since these results may not be replicated with more objective or independent measures (Reiter-Palmon et al., 2012). Self-assessment is prone to bias. In the future, a method that combines self-evaluation and instructor evaluation should be adopted. Third, this study revealed the mechanism through which collective thriving affects team creativity but did not explore the possible boundary conditions in this relationship. According to Csikszentmihalyi's (1999) theory of creativity, humans and their environment interact to produce

creativity. Future studies should thus consider how potential environmental factors (e.g., social networks and instructor feedback) moderate the process through which collective thriving influences team creativity. Our results revealed the mediating role of team voice and the team mental model, but other variables may also play mediating roles between collective thriving and team creativity. In teams with high psychological safety, members have interpersonal beliefs of mutual trust and respect; they tolerate mistakes, seek help and feedback, take risks, and offer different perspectives (Roussin et al., 2016). Team psychological safety is an important team process variable that promotes innovation behavior (Zhu et al., 2019). Therefore, future studies should consider other potential mechanisms that link collective thriving and team creativity (e.g., team psychological safety).

## 6. Conclusion

This study provides mechanistic insights into the association between collective thriving and college students' team creativity. We revealed that collective thriving was associated with team creativity. Furthermore, college students' innovation teams with higher-level collective thriving styles had greater team voice and a greater team mental model, which in turn contributed to their creativity. Thus, collective thriving contributes to college students' team creativity through the serial mediation of team voice and the mental model. Overall, this study reveals important findings related to collective thriving and team creativity that may be utilized to promote college students' team creativity.

## Ethical approval

All procedures performed in the study involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments. The study has been approved by Ethics Committee of Henan Normal University.

## Informed consent

Informed consent was obtained from all individual participants included in the study.

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## CRediT authorship contribution statement

**Hui Zhao:** Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Zhenzhen Wang:** Writing – review & editing. **Mengjiao Han:** Data curation. **Ying Huang:** Data curation.

## Declaration of competing interest

The corresponding author states that there is no conflict of interest.

## Data availability

Data will be made available on request.

## Appendix

### Collective thriving scale

1. We find ourselves learning often when working in the team.
2. We continue to learn more as time goes by when working in the team.
3. We see ourselves continually improving when working in the team.
4. We are not learning when working in the team (R).
5. We are developing a lot as person when working in the team.
6. We feel alive and vital when working in the team.
7. We have energy and spirit when working in the team.
8. We do not feel very energetic when working in the team (R).
9. We feel alert and awake when working in the team.
10. We are looking forward to each new day when working in the team.

### Team voice scale

1. Proactively develop and make suggestions for issues that may influence the team.
2. Proactively suggest new programs which are beneficial to the team.
3. Raise suggestions to improve the team's working procedure.
4. Proactively voice out constructive suggestions that help the team reach its goals.
5. Make constructive suggestions to improve the team's operation.
6. Advise other members against undesirable behaviors that would hamper team performance.
7. Speak up honestly with problems that might cause serious loss to the team, even when/though dissenting opinions exist.
8. Dare to voice out opinions on things that might affect efficiency in the team, even if that would embarrass others.
9. Dare to point out problems when they appear in the team, even if that would hamper relationships with other members.
10. Proactively report coordination problems in the team to the management (e.g., the instructor or the person in charge).

### Team mental model scale

1. If asked, I can clearly state what each person on my team does and how they work together.
2. Whenever I need it, team members will reach out to me and offer help or advice.
3. Team members know where to find the information they need.
4. I am aware of the skills and abilities of other team members.
5. I am good at guessing what other team members will do in certain situations.
6. Team members will adapt their own behavior to meet the needs of other members.

### Team creativity scale (creativity or innovation as a whole)

1. Team members often come up with new ideas.
2. Team members can implement new ideas smoothly.
3. Team members often find improved ways to do things.
4. Team members can create better processes and routines.

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